

INCITE report and renewal

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1 Requirements

1.1 Email contents from Zhenglu Li

Zhenglu Li Attachments Jun 10, 2026, 1:54 PM (13 days ago) to Weichen, Gurjyot, Shang, me, Jinyuan, Aaron, Emily, Stephen, Zui, Jimmy, Zien, Chih, Steven, Jeffrey, Diana, Felipe, Mauro, Zhenglu

Dear All,

Our INCITE project now requires the submission of a renewal proposal for the resources of the second year (2027) as the standard procedure. I would like to thank you in advance for your help and kindly ask for input for the renewal proposal, by June 26, Friday. If the point of contact in your group has changed, please kindly let me know.

Detailed instructions can be found here: <https://doeleadershipcomputing.org/proposal-renewal-instructions/>. At the end of the email, I will highlight some key instructions.

In preparing your input, please weigh Achievements vs. Plans for next year with a 2:1 ratio. In your input, please include 1. Project Achievements, 2. Project Plans for Next Year, 3. Publications Resulting from INCITE Awards, along with references.

I attach our proposal from last year for reference of proposed research and milestones. If you have any questions, please feel free to let me know. Thank you again for your very kind help and support!

Best, Zhenglu

1.2 More instructions

===== Key Information from Instructions on Renewal Proposal =====

Project Achievements: The Project Achievements should address the following points. This section is typically about 10 pages. Significance of Accomplishments to Date: Explain what advances you accomplished through the INCITE award (e.g., impact on community paradigms, valuable insights into or solving

a long-standing challenge). Place the proposed research in the context of competing work in your discipline or business. Reiterate the milestones of your proposal and discuss the accomplishments (planned or unplanned) achieved this year relative to those milestones and allocation use. Summarize the impact of the results achieved: What conclusions can be drawn and what is solved because of these results? What new and follow-on investigations have these results motivated? Allocation Use: Summarize your project's allocation use to-date this year: percent of allocated node-hours used on each platform, job size distribution, number of users, etc. Associate the resource use with particular results where possible. Also summarize your project's projected use from now until the end of December (i.e., end of current allocation year): anticipated percent of allocated node-hours used on each platform, job size distribution, etc. Associate this resource usage with anticipated results. Do you expect your usage to be evenly distributed throughout the remainder of this year? If not, explain. Application Parallel Performance: Summarize the performance (percent of peak, scalability) of your project's production application used in the allocations this year. What progress was made in improving the application's performance on this architecture? What challenges (if any) remain? List the technical risks and challenges that were confronted by your project (overcome or not) this year. Were they anticipated? Data Storage: What is the current cumulative size of stored data? What is your projection for the cumulative size of stored data at the end of the project? What tools and/or plans do you have to reduce the data? To share the data? The LCF data management policies can be found at: ALCF: <http://www.alcf.anl.gov/user-guides/data-policy> OLCF: <https://www.olcf.ornl.gov/for-users/olcf-policy-guide/>

Project Plans for Next Year: The Project Plans should address the following points. This section is typically about 4 pages. Summarize the Project Plans: Briefly explain what advances you expect to accomplish through the next award period and associate these with the overarching goals of your project. Clearly explain the relationships between the milestones, planned production simulations, and expected compute time required for these sets of simulations. Explain any change in the scope of the project (research objectives, computational approach, personnel, etc.) relative to the plans and approach articulated in the original proposal. If resource requirements differ from those of the previous year, provide details on the differences (platform, increased/decreased node-hours, file system and archival storage, networking) and the reasons for them. If you are requesting a new resource, you must provide evidence that your project is optimized to run on that resource. See the New Code Applications section below. Summarize the requirements that are driving the differences and what science/technology outcomes are expected. Significant changes to the original project scope should be discussed with the INCITE Program Manager prior to submittal. Development Work: Describe what, if any, development work has been carried out and the outcome of this work. Describe what additional development work will be executed, and when. Provide an estimate for the percentage of project time you will spend on development computing (i.e., porting, performance analysis) and other nonproduction runs. New Code Applications (where relevant): Are you planning to use any new production codes next year that were not included in your original proposal? Or are you proposing use of a new resource not included in your original proposal? If so, provide direct evidence, including supporting quantitative data, for your production application's parallel performance for the intended research simulations and analysis. Ideally, the proposing team will have generated the data. If you cite work by others, explain why it is applicable here. You should use the application code you intend for the production work, not a related code. Data for sample systems not related to the intended research is undesirable. Scaling and performance data in either strong or weak scaling mode should be provided. Explain how the strong or weak scaling applies to the proposed work. For data analytics and AI proposals, a description of the compute, memory, networking, and storage needs in the context of the LCF resources will help reviewers understand the strong or weak scaling characteristics. See the examples at the end of this page. NOTE: You may apply for a startup account at one of the centers to conduct performance studies. Director's Discretion applications are available at ALCF: <https://www.alcf.anl.gov/science/directors-discretionary-allocation-program> OLCF: <https://my.olcf.ornl.gov/project-application-new>

Publications Resulting from INCITE Awards: Provide a list of publications resulting from your current and previous INCITE awards to this project team for work related to the proposal under consideration. Only those publications that include an acknowledgement to INCITE and/or the LCF may be included. This list may not be used in lieu of references in Section 1, "Project Achievements and Plans." If applicable, list the citations in both Sections 1 and 2. (Does not count toward the 14-page Project Achievements and Plans limit).

2 Project Achievements

2.1 Significant Accomplishments to date

Things to answer:

- Advances accomplished.
- Impact on community paradigms.
- Valuable insights into some long standing challenge.
- Reiterate milestones of proposal and accomplishments (planned and unplanned).
- Impact of results achieved.
- What new and follow-on investigations have these results motivated.

TODO

2.2 Allocation use

Things to answer:

- Percent of allocated node hours used.
- Job size distribution.
- Number of users.
- Associate resource use with particular results if possible (can be less strict about this).
- Allocated node hours used.
- Possible usage and anticipated resource use with results.
- Resource use to be even throughout the year? Explain.

TODO

2.3 Application Parallel Performance

Things to answer:

- Summarize the performance (percent of peak, scalability) of your project's production application used in the allocations this year
- What progress was made on improving application's performance on this architecture.
- What challenges remain? Were they anticipated?

2.4 Data Storage

Things to answer:

- Cumulative data storage.
- Projection on cumulative data for future.
- Tools/plans to reduce data.

TODO

3 Project Plans for Next Year

3.1 Summarize the project plans

Things to answer:

- What advances I plan to accomplish through the next award period.
- Associate the above with overarching goals of project.
- If resource requirements differ for future, provide details.

TODO

3.2 Development work

Things to answer:

- Describe development work carried out as part of this project.
- Any additional development work if will need to be completed.
- Estimate of percent of project time spent on developing.

TODO

3.3 New Code applications

Things to answer:

- Any plans to use production code in the next year? Or new resource.
- Provide if new, the resource parallel info.
- Scaling and performance data in either strong or weak scaling mode should be provided.
- For data analytics and AI proposals, a description of the compute, memory, networking, and storage needs in the context of the LCF resources will help reviewers understand the strong or weak scaling characteristics.

TODO

4 Publications Resulting from this INCITE work

Things to answer:

- List Papers from this. Only ones with INCITE cited will be included. (I have none, only anticipated ones under preparation).

TODO